

Order Statistics

$\omega \in \Omega: X_1(\omega), X_2(\omega), \dots, X_n(\omega)$
 $\max\{X_1(\omega), X_2(\omega), \dots, X_n(\omega)\}$ a function from Ω to $\mathbb{R}$

Suppose $X_1, X_2, \dots, X_n$ are independent random variables with a common cdf F and density f .

- Let U denote the maximum of the X_i 's.

Then the cdf of U is given by

$$F_U(u) = P(U \leq u) = P(X_1 \leq u)P(X_2 \leq u) \cdots P(X_n \leq u) = (F(u))^n.$$

and so differentiating gives the density

$$f_U(u) = nf(u)(F(u))^{n-1}.$$

$$P\{\omega \in \Omega: \underbrace{U(\omega)}_{\max\{X_1(\omega), X_2(\omega), \dots, X_n(\omega)\}} \leq u\}$$

$$\begin{aligned} & P\left\{ \underbrace{X_1(\omega) \leq u, X_2(\omega) \leq u, \dots, X_n(\omega) \leq u}_{\substack{X_1(\omega) \leq u \\ X_2(\omega) \leq u \\ \vdots \\ X_n(\omega) \leq u}} \right\} \\ &= P\{\omega: X_1(\omega) \leq u\} \cdot P\{\omega: X_2(\omega) \leq u\} \cdots P\{\omega: X_n(\omega) \leq u\} \\ &= F(u) \cdot F(u) \cdots F(u) = (F(u))^n \end{aligned}$$

$$F_V(v) = P\{V \leq v\}$$

Likewise, let V denote the minimum of the X_i 's. Then

$$\begin{aligned} 1 - F_V(v) &= P(V \geq v) \\ &= P(X_1 \geq v)P(X_2 \geq v) \cdots P(X_n \geq v) = (1 - F(v))^n. \end{aligned}$$

Hence

$$F_V(v) = 1 - (1 - F(v))^n$$

and

$$\begin{aligned} f_V(v) &= (1 - (1 - F(v))^n)' = -n(1 - F(v))^{n-1}(-f(v)) \\ &= nf(v)[1 - F(v)]^{n-1}. \end{aligned}$$

Example

Suppose that $X_1, X_2, \dots, X_n \sim \text{Exp}(\lambda)$ are independent. Find the distribution of $\min\{X_1, X_2, \dots, X_n\}$.

By the previous slide, the density is given by

$$f_V(v) = n f(v) \cdot (1 - F(v))^{n-1}$$
$$f_V(v) = \underline{n}(\underline{\lambda e^{-\lambda v}})(\underline{e^{-\lambda v}})^{n-1} = \underline{n\lambda} \left(\underline{e^{-\lambda v} (e^{-\lambda v})^{n-1}} \right) = \underline{(n\lambda)} \underline{e^{-(n\lambda)v}}.$$

Thus $V \sim \text{Exp}(n\lambda)$.

$$\boxed{\lambda e^{-\lambda'v}} \\ \exp(\lambda')$$

Joint distribution of Min and Max

Recall that $U = \max X_i$ and $V = \min X_i$. Consider the joint CDF $P\{U \leq u, V \leq v\}$.

- When $u \leq v$,

$$U(\omega) \leq u \leq v \Rightarrow \underline{V(\omega)} \leq \underline{U}$$

$$F(u, v) \leftarrow P\{U \leq u, V \leq v\} = P\{U \leq u\} = (F(u))^n$$

- When $u \geq v$,

$$U(\omega) \leq u \begin{cases} U(\omega) \leq u, V(\omega) \leq v \\ U(\omega) \leq u, V(\omega) \geq v \end{cases}$$

$$F(u, v) \leftarrow P\{U \leq u, V \leq v\} = P\{U \leq u\} - P\{U \leq u, V \geq v\} \\ = (F(u))^n - (F(u) - F(v))^n$$

Hence the joint density function is

$$\frac{\partial F}{\partial u \partial v} f(u, v) = \begin{cases} 0, & (n f(v) (F(u) - F(v))^{n-1})'_u, & u \leq v; \\ n(n-1) f(u) f(v) (F(u) - F(v))^{n-2}, & u \geq v. \end{cases}$$

$$\begin{aligned} & u \leq u, V \geq v \\ & P\{v \leq V \leq U \leq u\} \\ & = P\{v \leq \begin{bmatrix} X_1(\omega) \\ X_2(\omega) \\ \vdots \\ X_n(\omega) \end{bmatrix} \leq u\} \stackrel{\text{by independence}}{=} P\{v \leq X_1 \leq u\} \cdots P\{v \leq X_n \leq u\} \end{aligned}$$

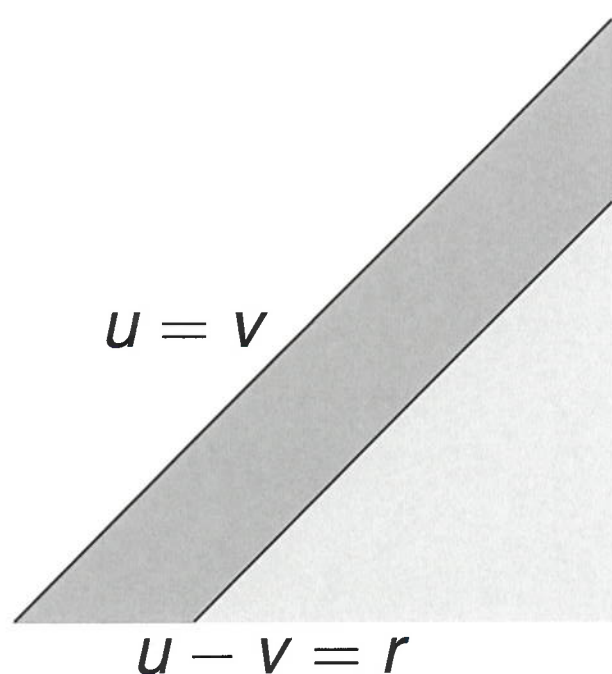
- If the X_i 's are uniform on $[0, 1]$, then

$$f(u, v) = \overbrace{n(n-1)}^{n(n-1) f(u) f(v)} (F(u) - F(v))^{n-2}, \quad 1 \geq u \geq v \geq 0.$$

Question

Let $R = U - V$ be the range of $X_1, \dots, X_n$. What is the density function of the range for uniform random variables?

Solution: Find the CDF of $R = U - V$ first and then differentiate it.



$$\begin{aligned}
 & F_R(r) \\
 &= \underline{P\{R \leq r\}} = \underline{P\{U - V \leq r\}} \\
 &= \int_{-\infty}^{\infty} \int_v^{v+r} \underline{f(u, v) du dv} \\
 &\quad \text{let } \boxed{u = v + s} \quad \underline{v \leq u \leq v + r} \\
 &\quad \quad \quad s = u - v \\
 &\quad \quad \quad 0 \leq s \leq r \\
 &= \int_{-\infty}^{\infty} \int_0^r \underline{f(v + s, v) ds dv} \\
 &= \int_0^r \int_{-\infty}^{\infty} \underline{f(v + s, v) dv ds}
 \end{aligned}$$

$$\frac{d}{dr} P\{R \leq r\} = \frac{d}{dr} \int_0^r \int_{-\infty}^{\infty} \underline{f(v + s, v) dv ds}$$

which gives

$$\begin{aligned}
 \underline{f_R(r)} &= \int_{-\infty}^{\infty} \underline{f(v + r, v) dv} \\
 &= \int_{-\infty}^{\infty} \underline{f(v + 0, v) dv} \quad // 0
 \end{aligned}$$

Joint density of (max, min) proved just now:

$$f(u, v) = n(n-1)(u-v)^{n-2}, \quad 1 \geq u \geq v \geq 0.$$

Then

$$f_R(r) = \int_0^{1-r} f(v+r, v) dv = n(n-1)r^{n-2} \underbrace{\int_0^{1-r} 1 dv}_{1-r}, \quad 0 \leq r \leq 1.$$

$0 \leq v \leq v+r \leq 1$
 $0 \leq v \leq 1-r$

MH2500 Probability and Introduction to Statistics

Slides for Week 10-1

Topics: Expectation of sums

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Expectation of Sums of Random Variables

In the case of a single random variable (discrete or continuous), we have

$$\mathbb{E}(g(X)) = \sum_x g(x)p(x), \quad \text{or} \quad \mathbb{E}(g(X)) = \int_{\mathbb{R}} g(x)p(x)dx.$$

A similar result holds for two random variables.

Proposition

If X and Y have a joint mass/density function $p(x, y)$ then

$$\mathbb{E}(g(X, Y)) = \sum_{x,y} g(x, y)p(x, y) \quad \text{or}$$
$$\mathbb{E}(g(X, Y)) = \iint_{\mathbb{R}^2} g(x, y)p(x, y)dxdy.$$

Corollary

$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ whenever $\mathbb{E}(X)$ and $\mathbb{E}(Y)$ exist.

Proof: We prove the corollary in the continuous case.

$$\begin{aligned}\mathbb{E}(X + Y) &= \iint (x + y)p(x, y) dx dy \\&= \iint xp(x, y) dx dy + \iint yp(x, y) dx dy \\&= \int x \left(\int p(x, y) dy \right) dx + \int y \left(\int p(x, y) dx \right) dy \\&= \int x p_X(x) dx + \int y p_Y(y) dy \\&= \mathbb{E}(X) + \mathbb{E}(Y)\end{aligned}$$

Corollary

If $X \geq Y$ then $\mathbb{E}(X) \geq \mathbb{E}(Y)$.

Proof:

$$\begin{array}{lcl} \underline{X - Y \geq 0} & \Rightarrow & \boxed{\mathbb{E}(X - Y) \geq 0} \Rightarrow \mathbb{E}(X) - \mathbb{E}(Y) \geq 0. \\ \begin{array}{l} X(\omega) - Y(\omega) \geq 0 \\ \text{for all } \omega \in \Omega \end{array} & & \begin{array}{l} \boxed{\int (x - y) p(x, y)} \\ \Downarrow \\ E(X) \geq E(Y) \end{array} \end{array}$$

Definition

Let $X_1, \dots, X_n$ be i.i.d. variables having distribution function F and expected value μ . Such a sequence of random variables is said to constitute a sample from the distribution F .

- The quantity $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is called the **sample mean**.
- $\mathbb{E}(\bar{X}) = \mu$, i.e., $\bar{X}$ is an unbiased estimator of μ .

Proof:

$$\mathbb{E}(\bar{X}) = \mathbb{E} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n} \mathbb{E} \left(\sum_{i=1}^n X_i \right) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) = \mu.$$

$$X_1 \sim \text{Geometric}\left(\frac{N-1}{N}\right) \quad E(X_1) = \frac{N}{N-1}$$

Observe that $X_i \sim \text{Geometric}\left(\frac{N-i}{N}\right)$ and thus

$$E(X_i) = \frac{N}{N-i}.$$

Therefore

$$\begin{aligned} \boxed{E(X)} &= 1 + E(X_1) + E(X_2) + \cdots + E(X_{N-1}) \\ &= \underline{1} + \frac{N}{N-1} + \frac{N}{N-2} + \cdots + \frac{N}{1} \\ &= N \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N} \right) \\ &\quad \boxed{\approx N \ln N.} \end{aligned}$$

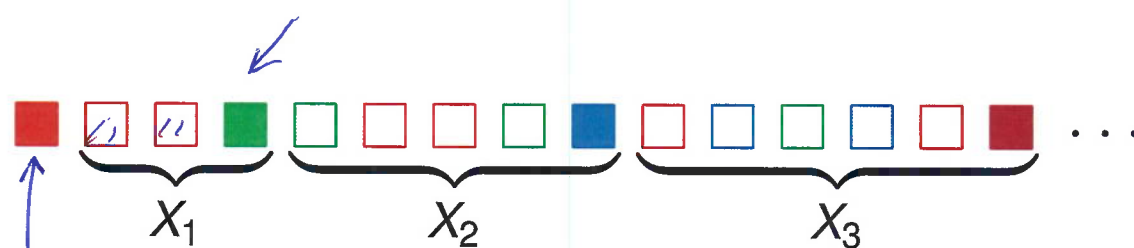
Recall the Euler's constant:

$$\lim_{N \rightarrow \infty} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N} - \ln N \right).$$

Coupon Collector Problem

There are N types of coupons. Each time one obtains a coupon, it is equally likely to be any one of the N types. Find the expected number of coupons one needs to amass before obtaining a complete set of at least one of each type.

Solution: Let X denote the number of coupons collected before a complete set is attained.



In general, X_i = the number of additional steps needed to see the $(i + 1)$ -st type after having collected distinct i types.

$$\mathbb{E}(X) = \underbrace{1}_{\mathbb{E}(X_0)} + \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_{N-1})$$

